National University of Computer and Emerging Sciences
Lahore Campus

μP Interfacing and Programming (EE3002)

Sessional-II Exam

Date: 4th November, 2025

Course Instructor(s)

Dr. Muhammad Akmal

Ms. Abeer Bashir

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 3

__________ Roll No   __________ Section   __________ Student Signature

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Instructions/Notes:
1. Attach the question paper with answer sheet.
2. Do not forget to write your Name and Roll Numbers.
3. Make assumptions if you think some data is missing.
4. Solve on the answer sheets. Answers written on the question paper will not be marked.

CLO 2: Develop a program to interface various peripherals with pins and ports of *microcontroller* **(PLO 3, C6)**

Q1: [7 marks]

Develop *a PIC18* assembly code snippet to copy the value 55H into RAM memory locations 40H to *42H using the following modes:*

i. Direct addressing mode.
ii. Register indirect addressing mode without a loop. INDF0 FSR0 L
iii. Register indirect addressing mode with a loop. POSTINC0

***CLO 1:* Describe** the internal architecture, programming of microprocessor and microcontroller **(PLO 1, C2)**

Q2: [7 marks]

a) **Explain** each of the following components of the internal memory architecture of the PIC18 microcontroller with a well-labeled comprehensive diagram:

I. 4K bytes of data memory
II. 2M bytes of program code memory
III. File Registers
IV. Banks
V. Special Function Registers (SFRs)
VI. General Purpose Registers (GPRs)
VII. Default Access Bank

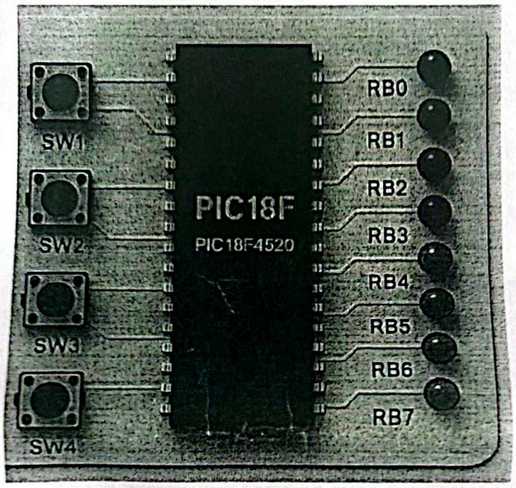
Note: Clearly label the respective sizes and address ranges of each of the above component.

CLO 3: **Develop** a program to interface various peripherals with pins and ports of microcontroller **(PLO 3, C6)**

Q3: [8+8 marks]

a) **Develop** an assembly language program that stores a 4-digit passcode in memory. The user will input the digits using switches connected to PORTD. The program should then compare the entered digits with the stored passcode. If all digits match, the LEDs on PORTB should turn ON; if they do not match, the PORTB LEDs should blink three times to indicate an incorrect passcode.

PORT D



b) **Develop** an assembly PIC18 code to concatenate arr_1 and arr_2, and store the result in arr_3.

Where, arr_1 = {1, 2, 0, 7, 0, 8} and arr_2 = {6, 7, 8, 8, 2, 9}

The result in arr_3 should be = {1,2,0,7,0,8,6,7,8,8,2,9}

LFSR0 POSTINC0

LFSR1 POSTINC1